AMERY ICE SHELF BOREHOLE MICROCAT AND ADCP DATA, 2010/2011 SEASON - DATA PROCESSING AND QUALITY

Mark Rosenberg (data processor)
Amery Ice Shelf borehole drill team (data collectors)

INTRODUCTION

This report summarises the latest AMISOR oceanographic data, season 2010/2011, collected from oceanographic equipment deployed through the Amery Ice Shelf boreholes. Time series data of temperature, salinity and pressure were downloaded from four strings of Seabird SBE37IM microcats at AMISOR boreholes AM03 (11 months of data), AM04 (23 months of data data), AM05 (13 months of data) and AM06 (12 months of data). Time series current profiles were obtained from an ADCP (RDI 300 kHz broadband Workhorse) on the AM06 string. Data processing and data quality details are included in the report.

Table 1: Station details for AMISOR season 2010/2011 borehole microcats and ADCP. Note that borehole absolute locations change in time, as the ice shelf is in motion. For the microcats, positive clock errors indicate microcat clocks were fast. Instrument pressures/depths are mean values from the pressure sensors over the data dates, including all pressure steps. For microcats 7327 and 7235, these means are after correction of pressure data. Note that data were not downloaded from microcats 1973 and 1974 at AM04 the previous season, so there's 2 years of data for these 2 instruments.

Borehole microcats

borehole	position at given date		microcat	instrument pos.		data dates		rec.	clock
	latitude	longitude		depth (m)	pressure (dbar)	start	end	interval (min.)	error (sec.)
AM03	70°32.680'S	70°20.470'E	4055	651.8	659.5	29-Jan-2010	22-Dec-2010	30	+145
AM03	(on 22-Dec-2010)		4054	859.6	870.2	29-Jan-2010	22-Dec-2010	30	+111
AM03	"	"	3883	1209.0	1225.0	29-Jan-2010	22-Dec-2010	30	+69
AM04	69°52.766'S	70°19.878'E	1972	518.4	524.4	04-Mar-2010	22-Dec-2010	30	?
AM04	(on 22-Dec-2010)		1973	661.9	669.7	30-Jan-2009	22-Dec-2010	30	?
AM04	"	"	1974	782.8	792.3	28-Jan-2009	22-Dec-2010	30	?
AM05	70°13.757'S	69°40.830'E	7327	573.8	580.5	19-Dec-2009	08-Jan-2011	30	?
AM05	(on 08-Jan-2011)		7326	730.0	738.8	19-Dec-2009	08-Jan-2011	30	?
AM05	"	"	7322	865.9	876.6	19-Dec-2009	08-Jan-2011	30	?
AM06	70°14.573'S	71°22.076'E	7325	545.1	551.4	07-Jan-2010	08-Jan-2011	30	+20
AM06	(on 08-Jan-2011)		7324	672.8	680.8	07-Jan-2010	08-Jan-2011	30	+29
AM06	"	"	7323	780.8	790.3	07-Jan-2010	08-Jan-2011	30	+14
AM06	"	"	ADCP	618.9	626.4	07-Jan-2010	08-Jan-2011	30	?

MICROCAT DATA

The microcats were downloaded by Adam Christensen (no field report available). The microcat moorings have already been described in previous field and data reports. This was the first download of data from the AM05 and AM06 instruments (deployed in December 2009 and January 2010 respectively). For AM03 and AM04 (deployed in December 2005 and January 2006 respectively), this was the fifth download. Note that at AM04 during the previous season (i.e. fourth download), data were only downloaded from microcat 1972, with no download from the microcats 1973 and 1974. So on this occasion 2 years of data were downloaded from microcats 1973 and 1974, all included in the data files for the fifth download.

All latest microcat data downloaded during the 2010/2011 season (Table 1) were assigned the same decimal time scheme consistent with previous data i.e. decimal days as counted from midnight on December 31st 2000 (so, e.g. midday on January 1st 2001 = 0.5 decimal time; midday on January 1st 2002 = 365.5).

The microcats were downloaded using the SeaBird terminal program Seaterm, and instrument clock errors were noted at the time for AM03 and AM06 (Table 1). Clock error information was lost for AM04 and AM05 instruments. From clock errors measured in previous seasons at AM04, the order of clock drift expected might be:

microcat 1972 = 120 sec fast (i.e. for the 2nd year of data from 04/03/2010 to 22/12/2010)
microcat 1973 = 380 sec fast (i.e. for the full 2 years of data)
microcat 1974 = 190 sec fast (i.e. for the full 2 years of data)

but note that these estimated corrections were not applied for AM04. For AM05, the instruments come from the same serial number run as AM06, and based on AM06 clock drifts a small drift only might be expected for AM05 instruments. For AM03, AM05 and AM06 microcats, microcat clocks were all reset to GPS time following data download and prior to reinitialisation of logging. For AM04, it is not known whether clocks were reset - the answer to this should be obvious from the magnitude of the clock error at the next download. For the ADCP at AM06, no clock check or clock set information was available.

Manufacturer supplied calibrations (June 2005 for AM03 instruments [except October 2004 for microcat-3883 pressure], February 2003 for AM04 instruments, and September 2009 for AM05 and AM06) were applied internally by the microcats, and calibrated data were output. The data were then processed, as follows:

* The files were padded at the start and end, data gaps were checked for and filled, and decimal times were calculated.

* Decimal times were "compressed" linearly throughout the time series to correct for clock error where available (Table 1).

* Mean pressure/depth values were calculated for the year's data for each microcat (Table 1), with all pressure steps included in the means.

Microcat data quality notes

A whole range of data features are present in the current season's download, from interesting real data features, to suspect but unexplained data, to obviously bad data.

The following bad data were removed from the files:

* AM03, microcat 4054 - conductivity/salinity data spikes removed for data points 299, 638, 5157-9, 8316-8328, 13304

* AM04, microcat 1972 – temperature/conductivity/salinity data removed for data points 24-59, due to slow ramping up of sensor data after instrument turned on

* AM04, microcat 1973 – conductivity/salinity data spikes removed for data points 29596, 32828

* AM04, microcat 1974 – conductivity/salinity data spikes removed for data points 739-740, 3067-3071, 11263, 12131-3, 14074, 19201

* AM05, microcat 7322 – conductivity/salinity data spikes removed for data point 93

* AM06, microcat 7323 – conductivity/salinity data spikes removed for data points 4953, 17376

* AM06, microcat 7325 – conductivity/salinity data spikes removed for data points 17487, 17491-4

Microcat data are plotted in Figures 1 to 5. From the pressure sensor data (Figure 5), iceshelf deformation events are evident at AM04 and AM05, as indicated by pressure steps. Strong current events, causing temporary "push-up" of the instruments, are largest at AM03, as indicated by pressure spikes (Figure 5).

For AM03, the pressure variation throughout the time series is very similar for all 3 microcats (Figure 5), with all pressures getting shallower by ~0.5 dbar over the first half of the times series. Although the order of this drift is of the same order as the expected pressure sensor drift as quoted by the manufacturer, the strong correlation in drift between the 3 instruments suggests that the drift is real.

For AM04, for microcats 1973 and 1974 there are 2 years of data from the current season's download. For microcat 1972, the instrument was successfully downloaded the previous season, so the current season's download only contains ~1 year of data. However the figures show the most recent 2 years of data from microcat 1972, for comparison to the other instruments. Note also that for microcat 1972, clock drift correction has been applied for data from 30/01/2009 to 04/03/2010 only; no correction was applied for data from 04/03/2010 onwards (no clock drift measurement was available, as discussed above).

Pressure data from AM04 (Figure 5) show significant vertical movement of the iceshelf, with good pressure correlation evident between the bottom 2 microcats. These pressure changes in the latter part of the record are not evident in microcat 1972 data, indicating a flattening of the batteries for this instrument. A further indication of suspect cat1972 data is the slow ramping up of sensor data at instrument switch on (04/03/2010), typical of flattening batteries. The conductivity step in early November 2010 (Figure 2) may also indicate failing batteries, although fouling could also be the cause. Clearly all microcat 1972 data from the latest download (i.e. 04/03/2010 onwards) are suspect, and should not be used.

For microcats 1973 and 1974 at AM04, pressure drifts over the time series correlate well, with a decrease of ~1 dbar before the large step, and a decrease of ~4 dbar for the 8 months after the step (Figure 5). This drift is likely to be real, particularly given the variability of the drift rate e.g. near the end of the record, the drift is very small.

For the shallowest microcat at both AM05 and AM06, there is a steady pressure drift over the year, ~13 dbar at for microcat 7327 at AM05, and ~7 dbar for microcat 7325 at AM06 (Figure 6). There is no evidence of this drift in the other microcats, and as such it cannot be real. Note that pressure sensitivity appears unaffected, as evidenced for example by the pressure step at AM05 around decimal day 3600, visible for all 3 microcats. One possible cause is icing of the pressure port, due perhaps to the proximity of the shallowest microcats to the water/shelf transition zone, and super-cooled water (particularly at AM05, from Figure 3) (discussed later in the report in the ADCP data section). In both cases, the pressure trend was removed from the data (assuming a linear trend), and conductivity/salinity values recalculated. From the results in Figure 5 (showing corrected data), there is a small variation from linear drift throughout the year, and as a result the corrected pressure for microcat 7327 may be in error by up to ~1 dbar; and corrected pressure for microcat 7325 may be in error by up to ~0.5 dbar. At the local temperature, conductivity and pressure values, this is equivalent to only a small salinity error, < 0.001 (PSS78).

From conductivity and temperature sensor calibration drift estimates in a previous data report for 2003/2004 data (i.e. 0.0003 (PSS78) **per month** for salinity, and 0.0005^{o}C **per year** for temperature), we might expect the following calibration drifts at the time of the latest 2010/2011 data downloads:

AM03 (in water for 5 years)	total S drift = 0.015 (PSS78)	total T drift = 0.0025^{o}C
AM04 (8 years since calibration)	total S drift = 0.029 (PSS78)	total T drift = 0.0040^{o}C
AM05/AM06 (in water for 1 year)	total S drift = 0.004 (PSS78)	total T drift negligible

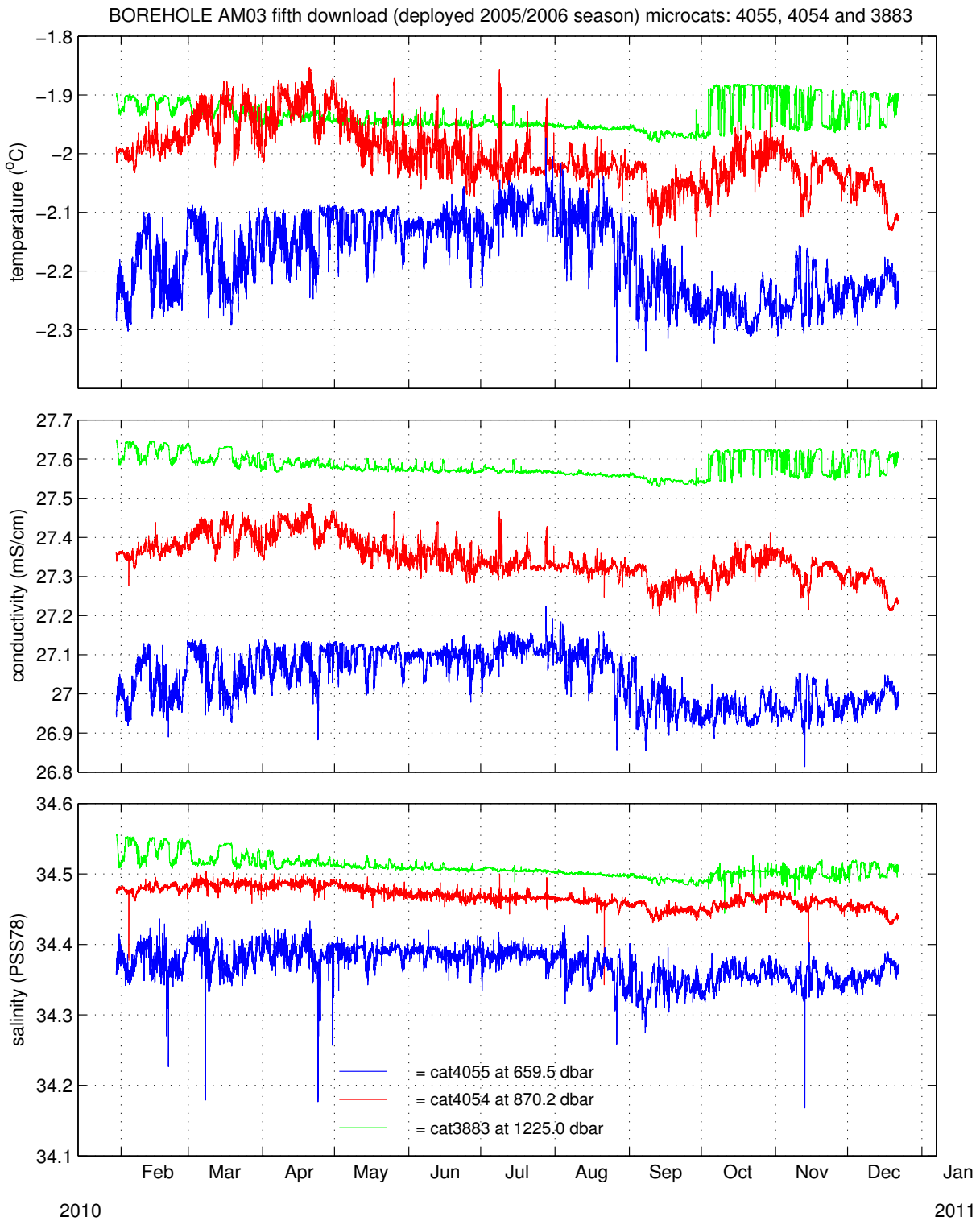


Figure 1: Borehole AM03 microcat data downloaded in 2010/2011 season.

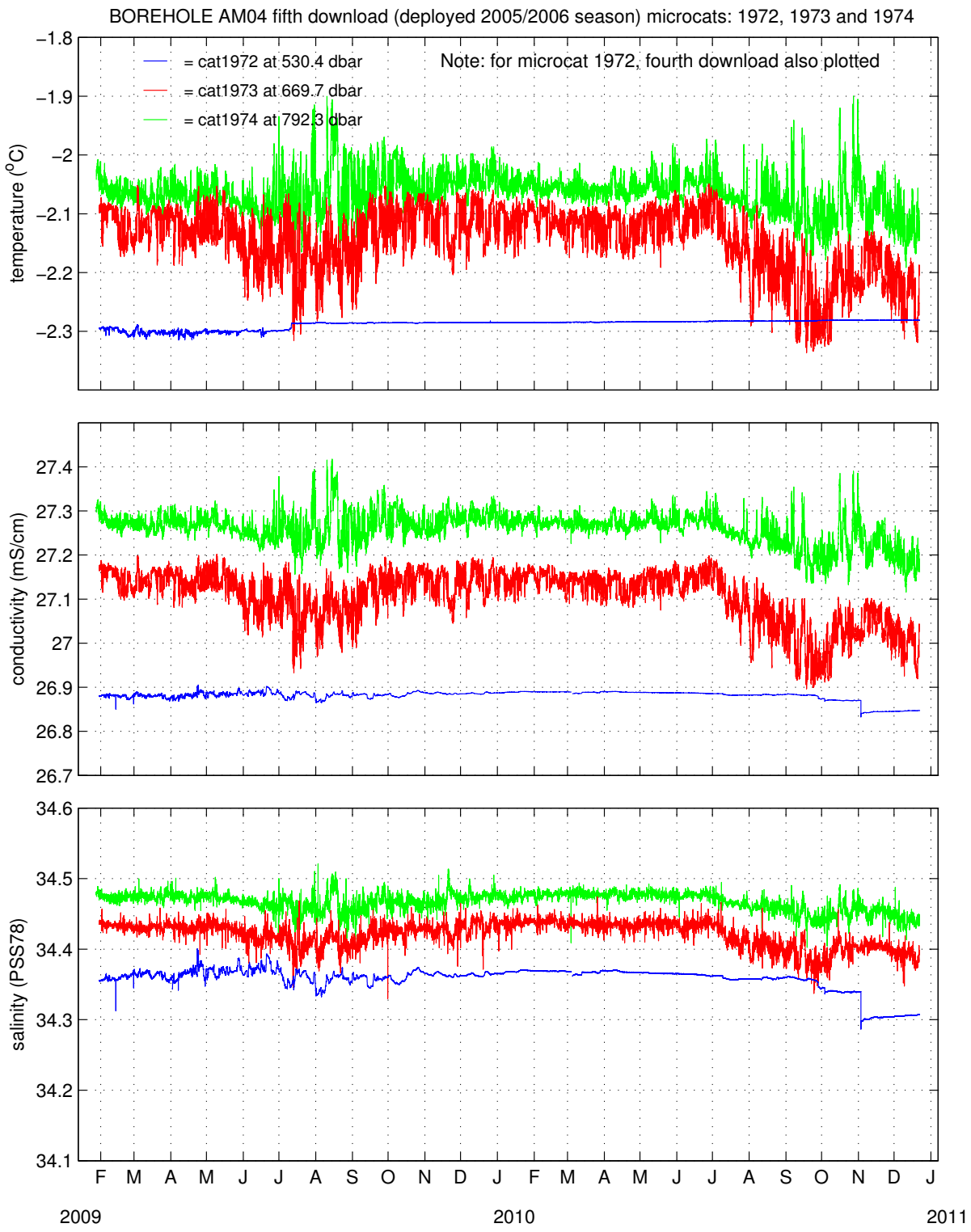


Figure 2: Borehole AM04 microcat data downloaded in 2010/2011 season – note that data for microcat 1972 up to 04/03/2010 is from the previous (i.e. fourth) download. Microcat 1972 data after 04/03/2010 are suspect.

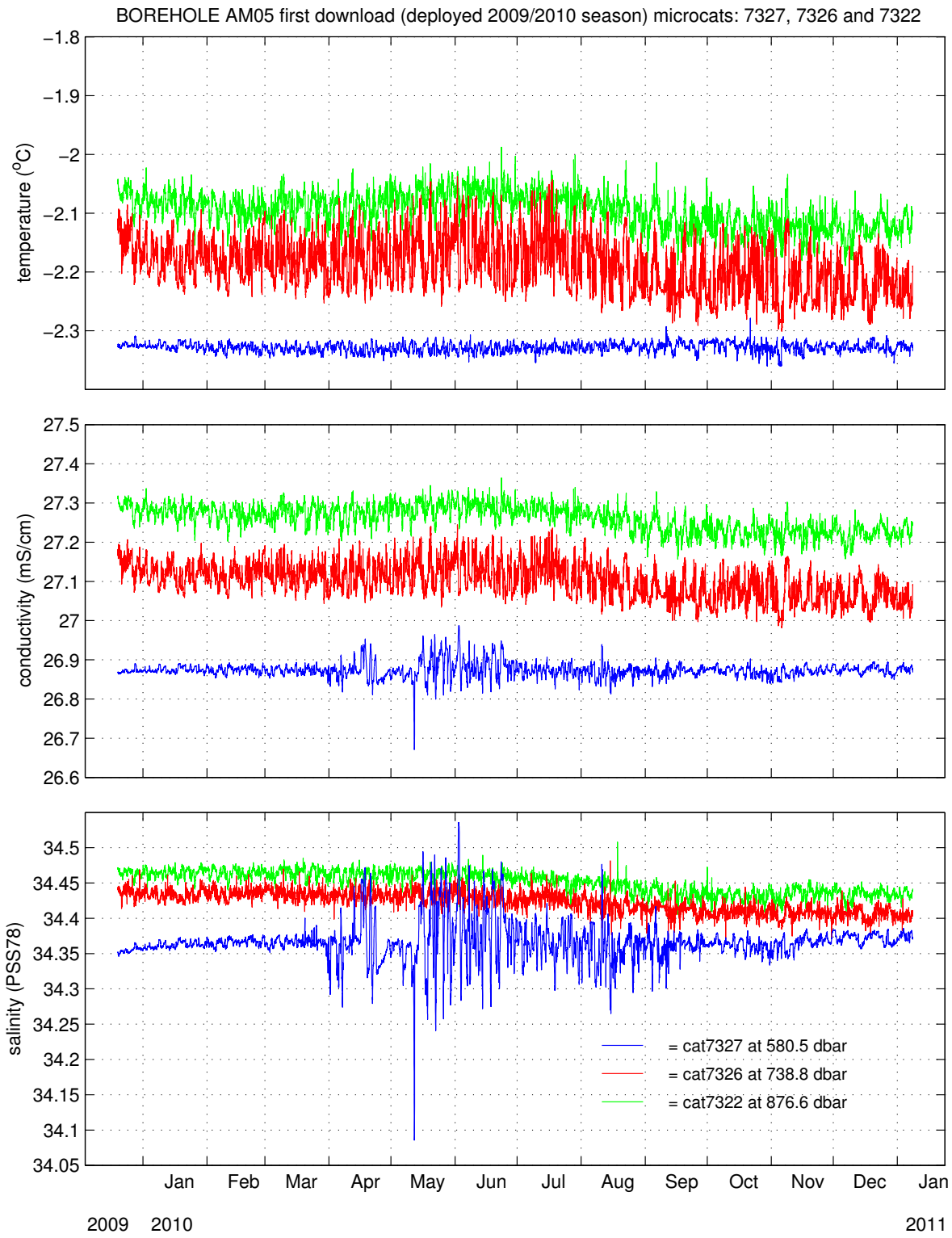


Figure 3: **Borehole AM05 microcat data downloaded in 2010/2011 season.**

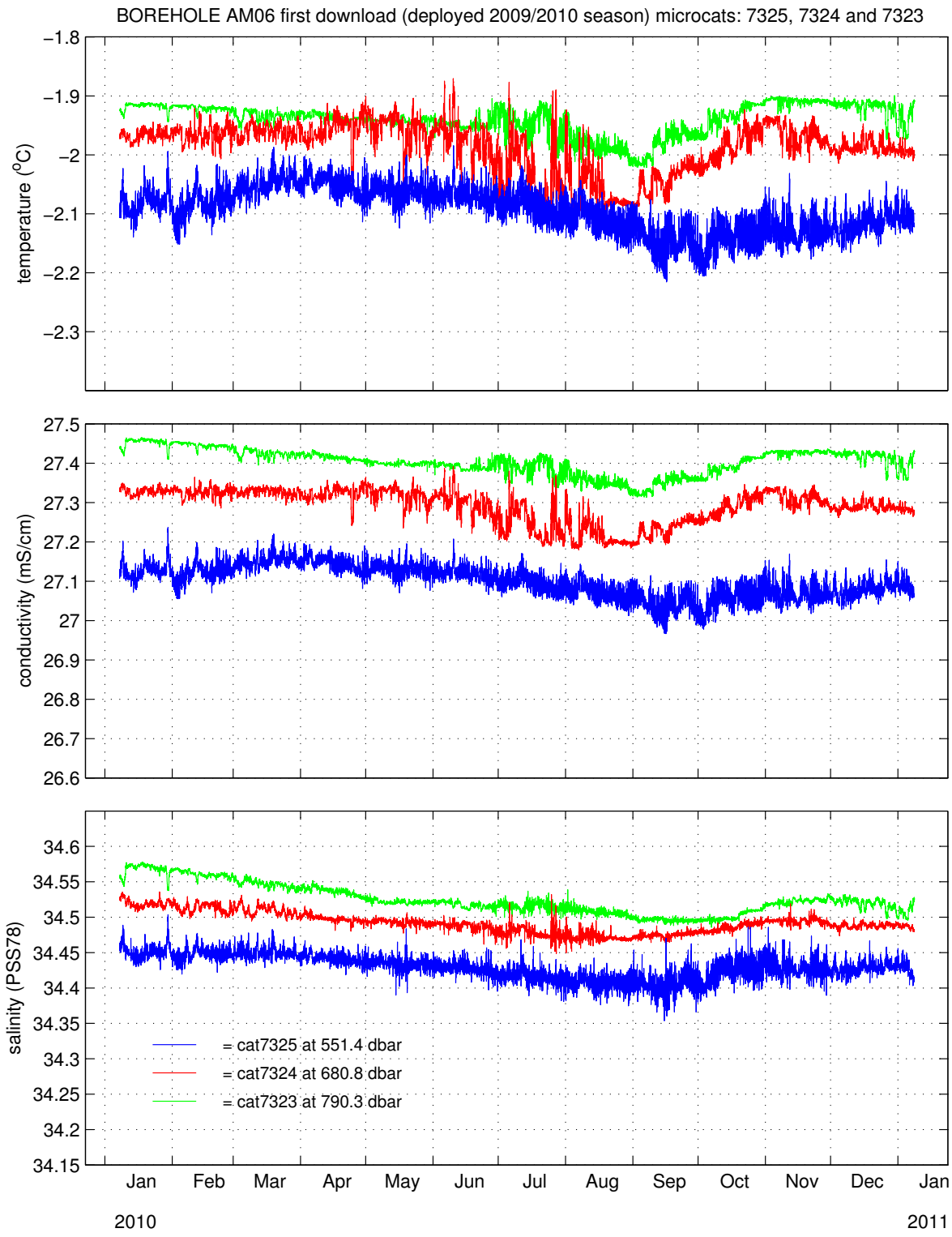
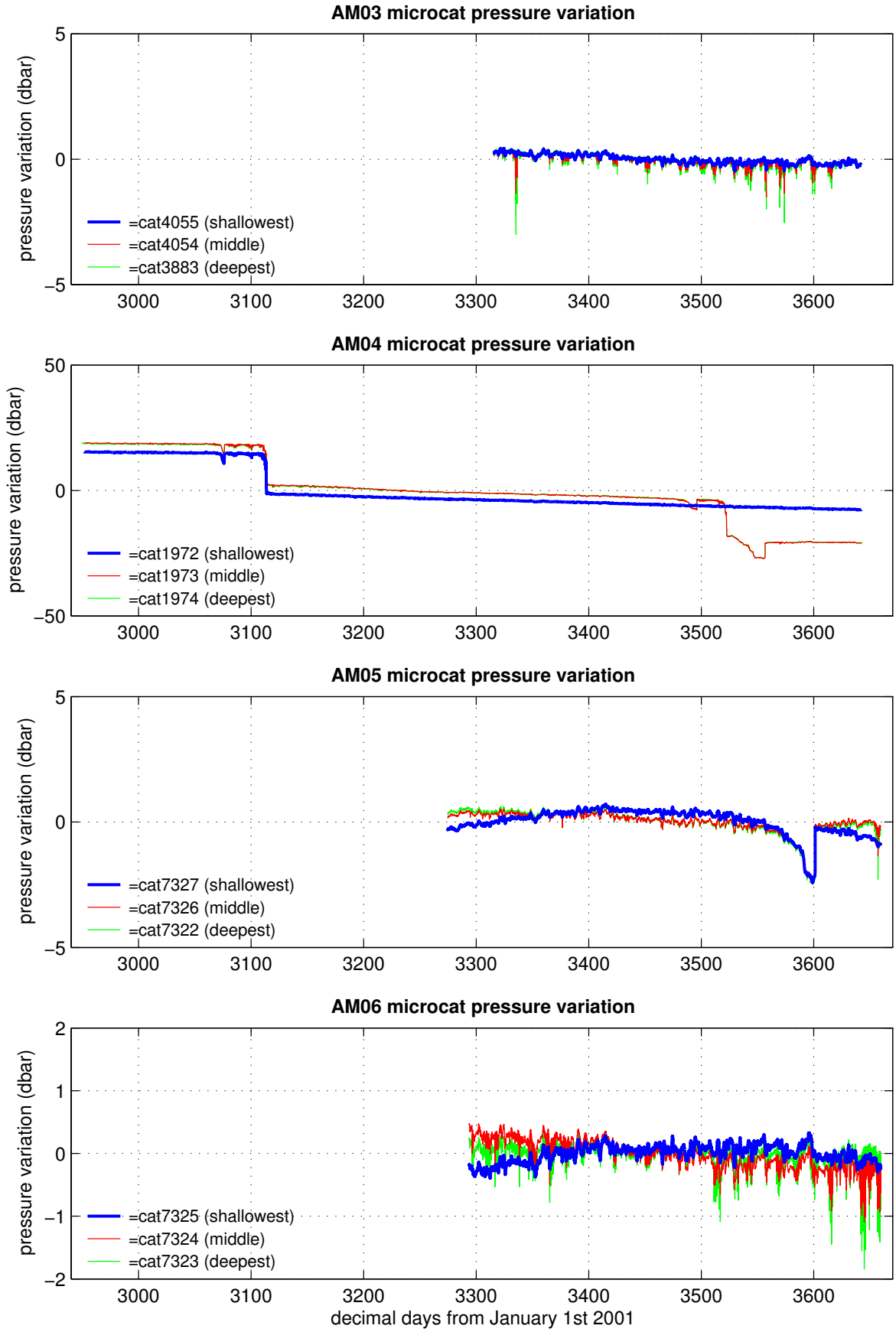


Figure 4: Borehole AM06 microcat data downloaded in 2010/2011 season.



<u>Figure 5:</u> Pressure sensor data from boreholes AM03, AM04, AM05 and AM06 microcats downloaded in 2010/2011 season. Plotted pressures are variations about the mean value. Note: pressures for AM05 and AM06 shallowest microcats are corrected.

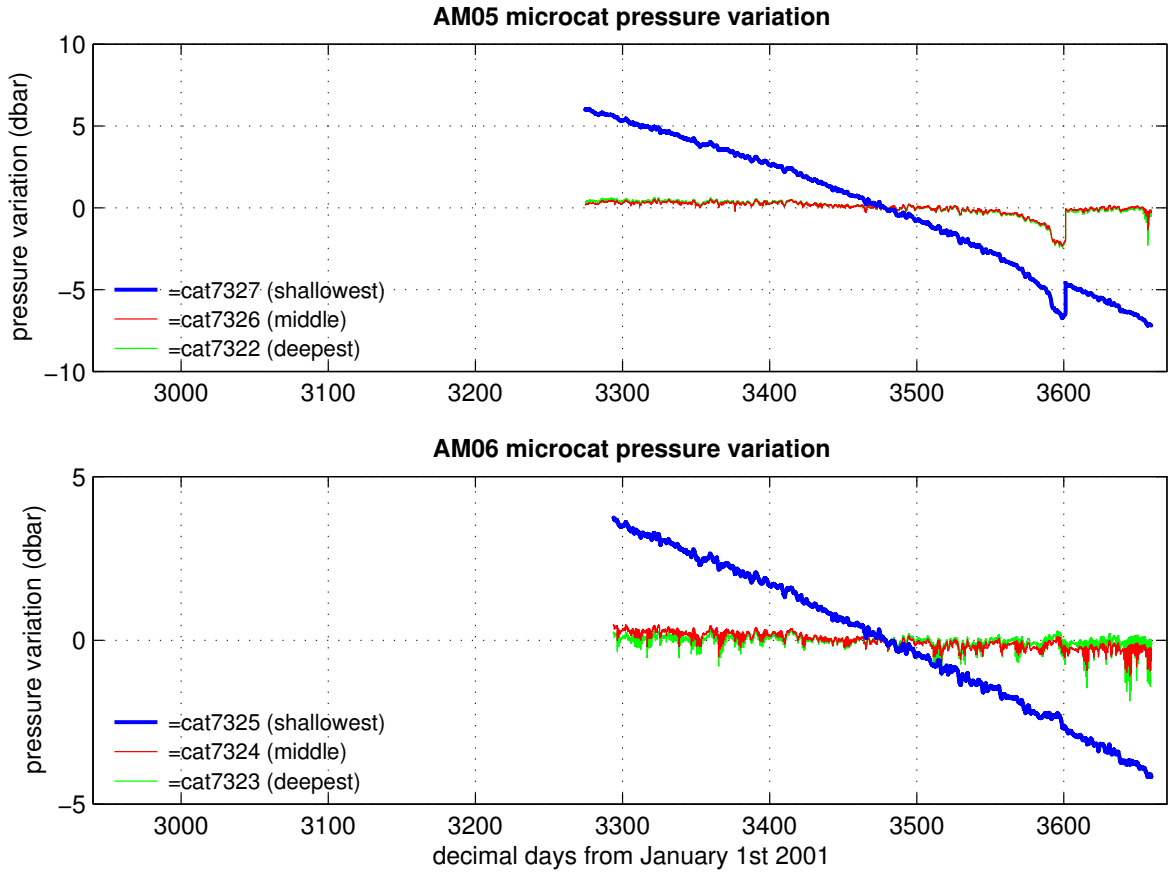


Figure 6: Pressure sensor data from boreholes AM05 and AM06 microcats, showing uncorrected pressures for shallowest microcats. Plotted pressures are variations about the mean value.

ADCP DATA

ADCP instrument and set-up details are as follows:

RDI Broadband Workhorse 300 kHz, 4 beam Janus configuration, beam angle 20 degrees
number of bins = 27
bin length = 4 m
seconds per bin = 0.8
pings per ensemble = 40
bottom track pings per ensemble = 1
reference layer = bins 1 to 5h
heading bias = -76°
coordinate transformation (EX) = 11111 i.e. data in earth coordinates, tilt corrections applied, 3 beam solutions allowed, bin mapping used
blank = 1.76 m
lag length = 0.47 m
transmit length = 4.20 m
→ middle of first bin at 6.09 m from transducers
ensemble interval = 30 min
sensors: pressure, heading, pitch, roll, temperature
deployment: transducers looking upward

The RDI program WinADCP was used to export data from the raw files into matlab, with data in engineering units. A matlab routine was then run to change the format of matlab matrices and vectors, calculate decimal time (referred to the same time base as for the microcats i.e. decimal days as counted from midnight on December 31st 2000), replace null values with NaN, convert all direction/attitude/temperature parameters to degrees, convert all speed/velocity parameters to cm/s, and convert all depth and bottom track values to metres. Note that there were no data gaps. Also note that no clock check or clock set information was available. All ADCP parameters are averaged over each 30 minute averaging period, and in the final data files the time assigned to each record is at the ***start*** of the averaging period.

Next, quality control was applied to the data, as follows. Speed, direction, and u/v/w component bin values are flagged as bad (i.e. replaced with NaN) when:

* pcntgd4 < 80% (where pcntdgd4 = percent good of 4 beam solutions used in making the ensemble);
* average beam correlation < 70;
* orientation flag = 0 (no cases of this found in the data);
* bins 27, 26, 25, 24 and 23 all flagged as bad, as discussed below.

Additional notes:

* bottom track distance for the first 7 ensembles flagged as bad

* In the final matlab files, rows 1 to 27 in all matlab matrices and vectors correspond to vertical bins 27 to 1 (i.e. row 27 = bin 1, which is the deepest bin for an upward looking ADCP).

* Instrument serial number is unknown, and in the final matlab files all matrices and vector names are prefixed with “111”, as a default instrument serial number

* ADCP battery data indicate rapidly dropping voltages towards the end of the time series, and it’s unlikely there’ll be enough internal battery capacity remaining for continued instrument operation.

* From pitch and roll data, the ADCP appears to be ~1° off vertical.

* Ping quality parameters (e.g. percent good and echo amplitude) indicate varying vertical measurement range throughout the year, most likely due to a seasonal variation in water column reflectors e.g. measurement range around ensemble 12000 is higher than at the start and end of the time series.

The instrument was inherited from the Chinese program in the early 2000’s - the calibration history prior to that is unknown, and no calibrations of auxiliary sensors have been done since. So sensor calibrations are at least 10 years old, and temperature and depth data should only be used qualitatively. Comparing ADCP depths (average ~619 m) to the expected depth as per the mooring diagram (634 m), ADCP depths appear to be ~15 m too shallow. When first deployed, the instrument was polled and a pressure of 640 dbar (equivalent to ~633 m) was noted, consistent with the mooring diagram. However the final depth data as converted by the WinADCP program appear to be in error. The correct density was used in the conversion, so the reason for the discrepancy is unknown. In the final data files, all bin depth values have been referred to the assumed instrument depth of 634 m. And note that the depth parameter error does not affect any other parameters.

Some ADCP profile data are plotted in Figure 7, including water speed, percent good (i.e. average percentage of 4 beam solutions used in making the bin), average beam correlation for the 4 beams, and average echo amplitude for the 4 beams. Note that the speed data plotted are prior to quality control. From the figure, at bin 23 (at ~540 m) no speed values are measured, and percent good values are equal to zero. There appears to be a physical feature at this bin level, either an ice crystal/slush layer or a sharp density surface, and all ADCP data above this layer (bins 23 to 27) are bad. This is confirmed by the rapid rise of beam correlation and echo amplitude counts in the top few bins (Figure 7), indicating contaminated bin data. This layer, at ~95 m from the ADCP, can be seen in the bottom track data (Figure 8), where both the ice shelf underside (as confirmed by a caliper measurement through the borehole) at 110 m above the ADCP, and the mystery layer at 95 m above

the ADCP, are evident. Note that the 110 m bottom track return for the ice shelf underside provides verification of the assumed ADCP depth of 634 m.

Earlier in the report, drifting pressure data were noted for the shallowest microcat at AM06. The microcat is situated just below the 95 m mystery layer, and this may be the cause of the pressure error. It's unlikely that an ice crystal layer is causing the pressure drift, as simultaneous fouling of the conductivity cell would be expected, however super-cooled water around this layer may somehow affect the pressure sensor stability.

The local magnetic declination correction for the year of data measurement at AM06 is -75.97^{o} (calculated using using the International Geomagnetic Reference Field IGRF11, as modelled by the International Association of Geomagnetism and Aeronomy Working Group V-MOD, and using the program geomag70 downloaded from the NOAA website www.ngdc.noaa.gov). The instrument data headers and set-up files contain conflicting information as to whether this correction was applied at the time of data logging, and therefore no correction was applied during the post processing. Data users need to make an independent check of current direction against other data sources (e.g. local CTD data or climatology) to determine whether the local magnetic correction needs to be applied or not.

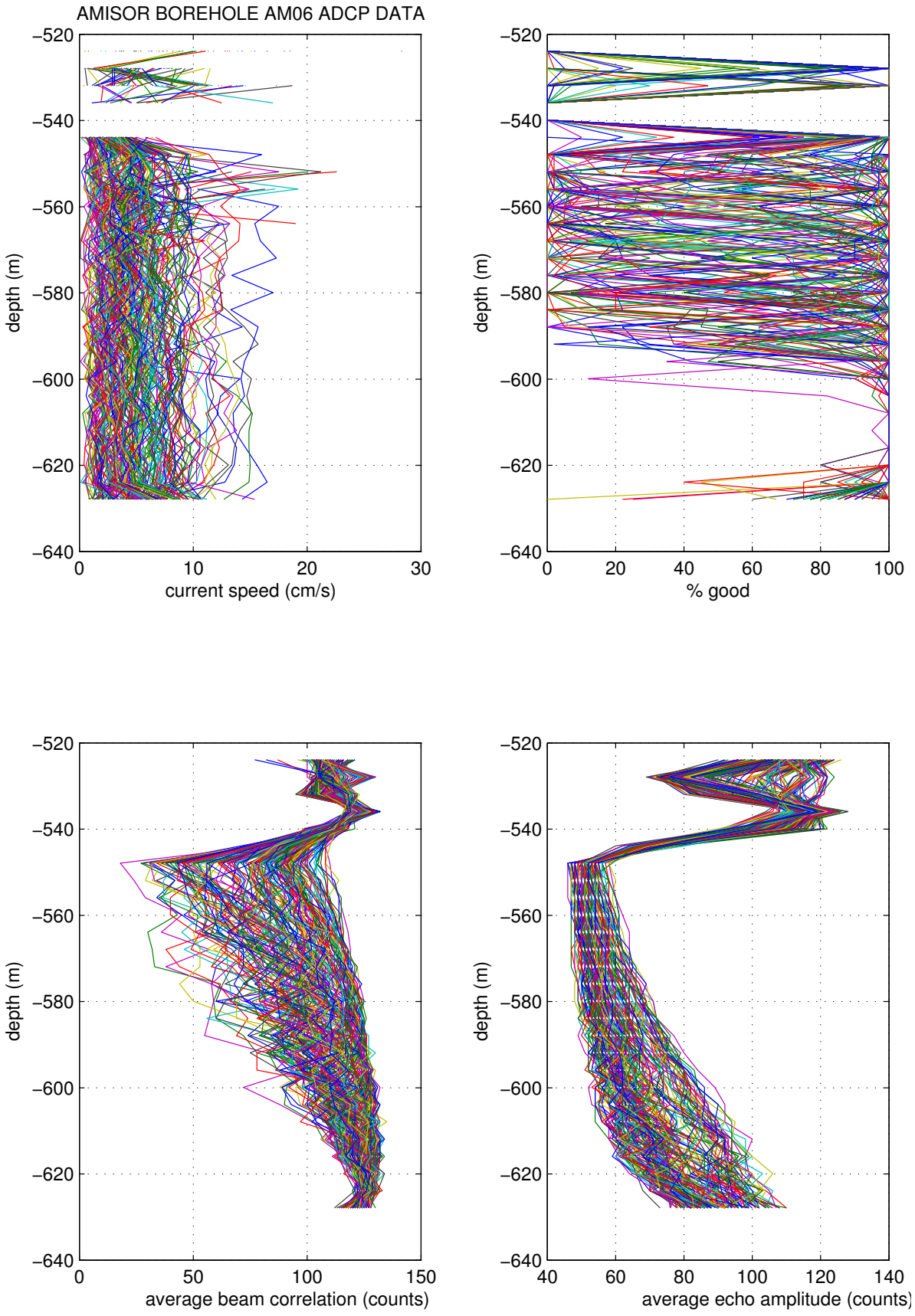


Figure 7: ADCP data from borehole AM06. Note that only every 100th data ensemble is plotted.

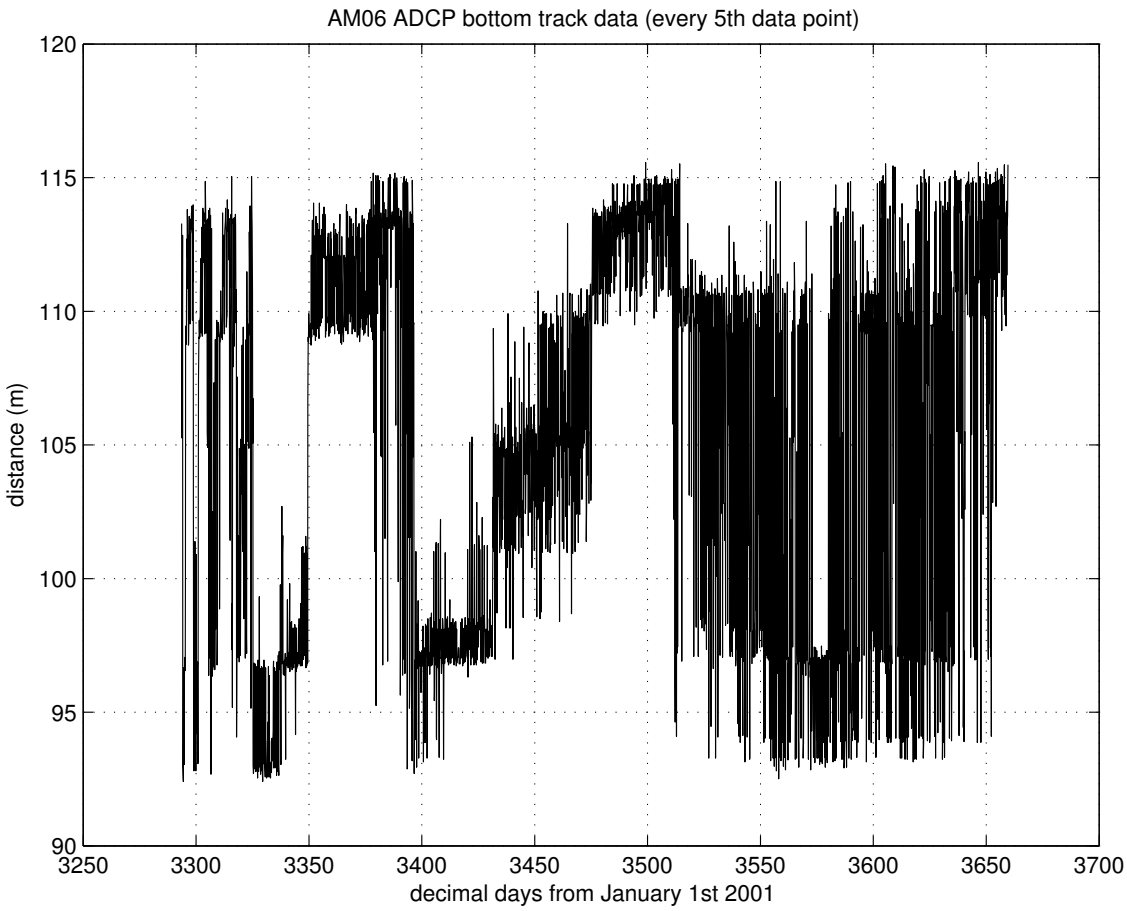


Figure 8: "Bottom track" ping data from ADCP at borehole AM06. Note that only every 5th data point is plotted. The "distance" shown is the distance from the ADCP transducers (note that ADCP is at water depth 634 m).

DATA FILES

New microcat data from boreholes AM03, AM04, AM05 and AM06, and ADCP data from borehole AM06, are contained in text and matlab format files, as follows:

1. Microcat data

AM03

Data from the fifth data download (i.e. 29/01/2010-22/12/2010) are contained in the files:
cat4055_5.dat, cat4054_5.dat, cat3883_5.dat (text files), and
cat4055_5.mat, cat4054_5.mat, cat3883_5.mat (matlab files).
READMEcatam03_5 details the microcat file formats.

The updated unified files (i.e. all data from initial deployment time on 17/12/2005, updated with data from the fifth download) are contained in the files:
cat4055_all.dat, cat4054_all.dat, cat3883_all.dat (text files), and
cat4055_all.mat, cat4054_all.mat, cat3883_all.mat (matlab files).
READMEcatam03_all details the microcat file formats.

AM04

Data from the fifth download (i.e. 04/03/2010-22/12/2010 for cat1972, 30/01/2009-22/12/2010 for cat1973, and 28/01/2009-22/10/2010 for cat1974) are contained in the files:
cat1972_5.dat, cat1973_5.dat, cat1974_5.dat (text files), and
cat1972_5.mat, cat1973_5.mat, cat1974_5.mat (matlab files).
READMEcatam04_5 details the microcat file formats.
The additional files cat1972_4_5.dat and cat1972_4_5.mat contain data from the fourth and fifth downloads, for ease of comparison to the other 2 microcats at AM04.

The updated unified files (i.e. all data from initial deployment time on 11/01/2006, updated with data from the fifth download) are contained in the files:
cat1972_all.dat, cat1973_all.dat, cat1974_all.dat (text files), and
cat1972_all.mat, cat1973_all.mat, cat1974_all.mat (matlab files).
READMEcatam04_all details the microcat file formats.

AM05

Data from the first data download (i.e. 19/12/2009-08/01/2011) are contained in the files:
cat7327_1.dat, cat7326_1.dat, cat7322_1.dat (text files), and
cat7327_1.mat, cat7326_1.mat, cat7322_1.mat (matlab files).
READMEcatam05_1 details the microcat file formats.
Data in these cat7327 files are corrected for the pressure drift.
Uncorrected data are in cat7327_1uncorrected.dat and cat7327_1uncorrected.mat

AM06

Data from the first data download (i.e. 07/01/2010-08/01/2011) are contained in the files:
cat7325_1.dat, cat7324_1.dat, cat7323_1.dat (text files), and
cat7325_1.mat, cat7324_1.mat, cat7323_1.mat (matlab files).
READMEcatam06_1 details the microcat file formats.
Data in these cat7325 files are corrected for the pressure drift.
Uncorrected data are in cat7325_1uncorrected.dat and cat7325_1uncorrected.mat

2. ADCP

AM06

Data from the ADCP at borehole AM05 are contained in the files:
adcp_am06_1qc.mat matlab file with the quality controlled data
adcp_am06_1noqc.mat matlab file with the complete data (including bad data) i.e. no quality control applied
READMEadcpam06_1 a README file detailing file formats

3. Station and summary info files

The files station.dat and borehole_data_summary.dat are updated to include the latest season's data. Note that for the shallowest microcats at AM05 and AM06 (i.e. serials 7327 and 7325 respectively), the mean depth values in these station files are after correction of the pressure sensor data.